

Deeply Exploit Visual and Language Information for Social Media Popularity Prediction

Jianmin Wu*
Alibaba Group
Hangzhou, China
jianmin.jm@alibaba-inc.com

Liming Zhao*
Alibaba Group
Hangzhou, China
lingchen.zlm@alibaba-inc.com

Dangwei Li*
Alibaba Group
Hangzhou, China
dangwei.ldw@alibaba-inc.com

Chen-Wei Xie
Alibaba Group
Hangzhou, China
eniac.xcw@alibaba-inc.com

Siyang Sun
Alibaba Group
Hangzhou, China
siyang.ssy@alibaba-inc.com

Yun Zheng
Alibaba Group
Hangzhou, China
zhengyun.zy@alibaba-inc.com

ABSTRACT

Social media popularity prediction task is to predict future attractiveness of new posts, which could be applied for online advertising, social recommendation, and demand prediction. Existing methods have explored multiple feature types to model the popularity prediction, including user profile, tag, space-time, category, and others. However, images and texts of social media posts, as important and primary information, are usually used by simple or insufficient processing. In this paper, we propose a method to deeply exploit visual and language information to explore the attractiveness of posts. Specifically, images are parsed from multiple perspectives including multi-modal semantic representation, perceptual image quality, and scene analysis. Different word-level and sentence-level semantic embedding are extracted from all available language texts including title, tags, concept and category. It makes social media popularity modeling more reliable with the powerful visual and language representation. Experimental results demonstrate the effectiveness of exploiting visual and language information by the proposed method, and we achieve new state-of-the-art results on the SMP Challenge at ACM Multimedia 2022.

CCS CONCEPTS

• **Information systems** → **Multimedia and multimodal retrieval**; • **Computing methodologies** → *Computer vision tasks*.

KEYWORDS

Social Multimedia, Visual Prediction, Popularity Prediction

ACM Reference Format:

Jianmin Wu*, Liming Zhao*, Dangwei Li*, Chen-Wei Xie, Siyang Sun, and Yun Zheng. 2022. Deeply Exploit Visual and Language Information for Social Media Popularity Prediction. In *Proceedings of the 30th ACM International*

*Equal Contributions

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from [permissions@acm.org](https://permissions.acm.org).

MM '22, October 10–14, 2022, Lisbon, Portugal

© 2022 Association for Computing Machinery.

ACM ISBN 978-1-4503-9203-7/22/10...\$15.00

<https://doi.org/10.1145/3503161.3551576>

Post (a)		Post (b)	
Image		Image	
Category	Travel & Active & Sports	Category	Urban
Title	Stretch Point Reyes, California	Title	IMG_2186
Tags	Sunset, Sea, Nature Landscape	Tags	Blackwhite Pattern Tiles
Popularity	16.42	Popularity	1.0

Figure 1: Samples of social media posts with visual and language content (e.g., image, category, title, tags and ground-truth popularity). Compared with post (b), we intuitively find that post (a) shows more beautiful image, attractive tags and lies in a popular category.

Conference on Multimedia (MM '22), October 10–14, 2022, Lisboa, Portugal. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3503161.3551576>

1 INTRODUCTION

The volumes of social media content increase super rapidly as uploading posts is a popular and easy activity on platforms such as Twitter, Facebook, Instagram and Flickr. Predicting the popularity of social media post is interesting and useful for many real-world applications, e.g., online advertising [19, 44], social media recommendation [17, 24, 28], topic miming system [9, 35, 42], fashion trending [12, 22], and information retrieval [13].

The task is challenging as we need to find the complicated clues of attractiveness for people from different multi-modal information including images, text, temporal date, spatial location, user profile and others. Previous methods [18, 38, 43] have explored many meaningful and significant discoveries on feature combination, importance weights and numerical processing. We find that images and texts are usually simply processed as categorical feature or single numerical feature vector, which is insufficient in our opinion since they are often direct information exposed for attracting users on social media platform. For example, a beautiful picture with hot topic tags is more likely to be popular than others with low-quality photos, as illustrated in Figure 1.

In this paper, we propose a method to deeply exploit visual and language information from the social media content for improving the performance of popularity prediction. We extract rich features from the pictures and texts uploaded by users on social media

platform, and try to explore the influence to the attractiveness or popularity of posts.

For the visual part, “A picture is worth a thousand words” is convincing especially on social media platforms, where people tend to share pictures and also like to view beautiful photos. An empirical study [21] shows that pictures with certain characteristics could be more attractive than some other pictures, such as image quality, the presence of face, and the semantic types (e.g., selfies, landscape photography, extreme sports). These information are self-contained in one picture uploaded by users, which is more reliable than other features. For example, the tags may be wrongly inputted by users by accident or on purpose for more exposure by platforms. Therefore, we propose to extract visual features from multiple perspectives including semantic representation, perceptual image quality, and scene analysis. We use raw feature vectors rather than simply tagging or computing cluster statistics, which enables discovering more sophisticated characteristics useful for social media popularity prediction.

For the language part, it provides another direct information to viewers and influence the attractiveness of the post. It is always true that a post with hot tags could obtain more views than others. Because some topics could appeal to a wide audience and some special-interest topics usually have limited popularity. Instead of only treating the tags or categories as categorical id features, we extract embedding from both the tags and categories. In this way, the language information could be more robust by using large language models than computing categorical features only from the training data, and we observe that converting categories to word embedding helps to improve the performance for popularity prediction.

In summary, we propose a method, called “DEVL”, to deeply exploit different visual and language information for social media popularity. We parse the picture from multiple perspectives to explore the characteristics for attractiveness, and extract semantic embedding from language texts instead of using categorical features. The experiments demonstrate that visual and language information really play an important role for popularity prediction task. The “DEVL” method achieves new state-of-art results on the Social Media Prediction challenge [40] at ACM Multimedia 2022.

2 RELATED WORK

Social Media Popularity Prediction. Social media popularity prediction is an interesting task for social media analysis, and attracts many researchers to explore the features or decision models [3, 4, 8, 10, 15, 18, 40, 41]. To explore effective features, existing methods use either hand-craft manner [4, 5, 10, 18, 37] or deep learning models [2, 8, 29] on feature extracting and fusion.

The hand-craft methods like CCP [5] and HyFea [18] usually construct multiple types of features including image, category, space-time, user profile, tag and others, and achieve a favorable predicting results. These features contribute a lot to the final performance, but it needs expert experiences and careful choices and sometimes is insufficient.

The deep learning methods like MFF [8] and DeepInf [29] trains a neural network to learn users’ latent social representation, which shows superior results of representation learning than traditional

hand-craft approaches. However, the learning is depended and influenced by the training data, and hard to adapt to all social media content.

Both the hand-craft and deep learning features are simplex or biased for the exploration on visual and language information. With the rapid improvement on multimodal learning and pretraining technology, we could learn more powerful and representative information beyond the limited social media data.

Multimodal Representation Learning. The multimodal representation learning model [14, 30], trained by a huge amount of paired image-text from the web, has scaled up visual-language pre-trained model to implement state-of-the-art results in zero-shot classification tasks. Some recent works begin to extend contrastive representation learning model to traditional vision tasks including image caption [6, 25], detection [11, 20, 34], video retrieval [1, 23], etc. In particular, generative models [26, 31, 32] based on CLIP [30] achieve results far exceeding previous works.

Considering that pre-trained models have strong cross-modal representation and achieved success in multiple domains, we introduce visual and language embedding extracted from multimodal models for Social Media Popularity Prediction.

3 METHOD

We deeply exploit different visual and language information for social media popularity on the basis of the last SMP Challenge [40] champion HyFea [18]. HyFea has completed a lot of feature engineering and data crawling, while exploring little in text and image deep processing. Take visual information as an example, only image length, width and other basic information are used. The proposed method will extract features of image and text in dataset from multiple perspectives. For the regression model, we adopt the same CatBoost [27] model as HyFea for simplicity and comparability. The overall framework of proposed method is shown in Figure 2.

3.1 User Information

The official SMP dataset contains some user information. We also follow the HyFea method to obtain more, including FollowerCount and TotalFaves, etc. More details can be found in [18]. For better geographic information use, we use Geocoder¹ to convert latitude and longitude into countries and cities, and then label countries and cities to get 3995 geographic categories. Meanwhile, we assume that the ranking of many user characteristics throughout the SMP dataset will also have a great impact on the popularity. So some user statistical characteristics are globally sorted, and the characteristics involved in sorting include uid count, all tags length, etc.

3.2 Visual Features

As mentioned earlier, the pre-trained contrastive representation model has achieved good generalization ability, so we use the image branch of the multimodal model trained by ourselves to extract the feature vector of the image. The model adopts a two-way framework similar to CLIP [30]. The difference is that our model uses a billion-level paired text-image dataset collected from public websites, then the text branch is changed to the BERT [7] structure, and the soft prompt is added to the input word embedding. After obtaining the

¹<https://geocoder.readthedocs.io>

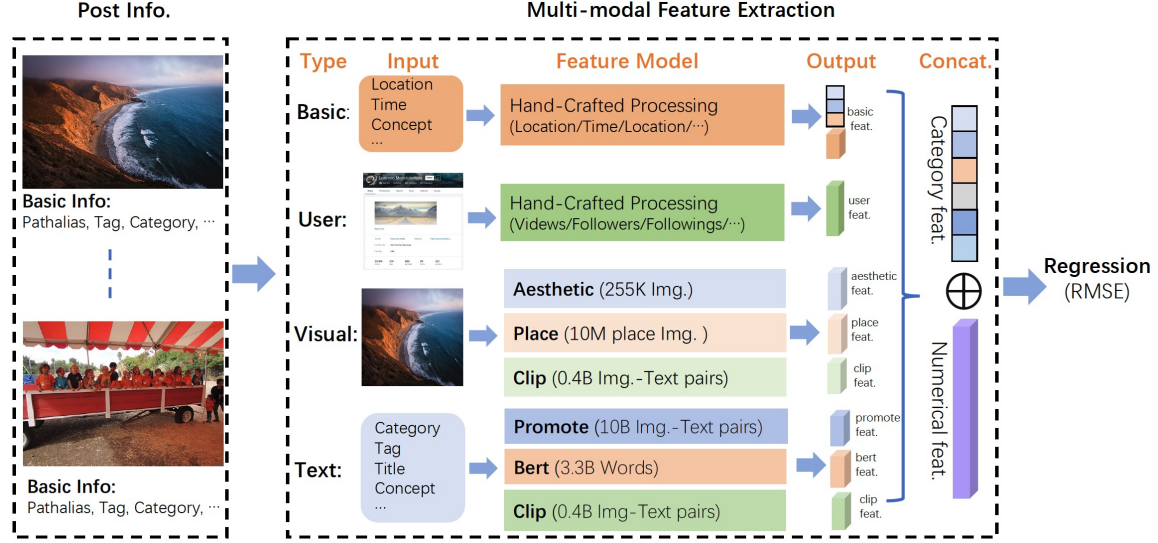


Figure 2: The overall framework of proposed method. Given the basic posts, four types of features are extracted, including basic post feature, user feature, visual features and text features from large-scale pretrained models. Finally, the concatenation of category and numerical features are utilized to train regression model of CatBoost using RMSE loss.

512-dimensional image feature vector, we use Principal Component Analysis(PCA) to reduce the dimension to 8 and 32, and finally concatenate all the vectors as input.

For the social media prediction of post, the quality of the image also play a big role. The image quality assessment (IQA) is to enable computers to perceive image quality like humans. Therefore, we use Nima [36] and HyperIQA [33] to analyze the image. Nima score images reliably and with high correlation to human perception, while HyperIQA has the ability of assessing image in the wild. We obtained assessment scores and corresponding variance.

We also believe that the image background has an influence on the popularity of the post. For example, the seaside background attracts more viewers' attention, so we use the Place365 [45] to obtain the image background label.

3.3 Text Features

Participants in previous competitions usually take text in SMPD as labels, so many of the original semantic of the text is lost. Here we extracted semantic of multiple characteristics including tags, category, etc.

In order to enhance the diverse representation of the same text, we simultaneously use multiple text models to extract vectors. The first uses the text branch of the self-trained prompt-enhanced multimodal model to obtain 512-dimensional feature vectors, and then use PCA to reduce the dimension to 8. Since most of text in the self collected dataset is in Chinese, there may be shortcomings in the representation of English text, so we also use Clip which has released multiple pre-trained models with different network structures, we choose a transformer-based text branch with 60 million parameters to extract 1024-dimensional feature vectors. We also use BERT model which consists of 12 Layer transformer composition from HuggingFace[39], pretrained on a large corpus of English data in a self-supervised fashion.

After concatenating all the features, the dimension of our training data finally reaches 8496 dimensions, far exceeding the previous champion.

3.4 Regression Model

Different from classification, generation and other tasks, regression task uses tree-based gradient boosting algorithm rather than deep model. As a gradient boosting toolkit, CatBoost implements the ordered boosting, a permutation-driven alternative to the classic algorithm, and adopts an innovative algorithm for processing categorical features leading to excellent results in the regression task.

4 EXPERIMENTS

In this section, the typical dataset that adopted is introduced firstly. Then two typical evaluation metrics are described. Finally, detailed experimental analysis of variant multi-modal features are conducted.

4.1 Datasets

The Social Media Prediction Dataset (SMPD) [40] is used to validate the effective of proposed idea. It contains 486K social multimedia posts which come from 70K users from Flickr and span about 16 months. Diverse social media information is collected in this dataset, including user profile, web image, post tags, time, category, anonymous photo-sharing records, etc. It is a multi-faced, large-scale, temporal web data collection dataset. Following official train/test splits, there are 306K posts for training and 180K posts for test. Without special instructions, experimental results on test split are reported in the following subsections.

4.2 Evaluation Metric

Following paper [40], two popular metrics are adopted to measure the algorithm performance, including Spearman's Rho (**SR**) and Mean Absolute Error (**MAE**). Specifically, SR is a non-parametric measure of rank correlation, which focus on ranking relevance between ranked predicted popularity scores set $\hat{P}$ and ground-truth popularity set P , varying from 0 to 1. For example, if there are N samples, the SRC can be expressed as follows:

$$SRC = \frac{1}{N-1} \sum_{i=1}^N \left(\frac{P_i - \bar{P}}{\sigma_P} \right) \left(\frac{\hat{P}_i - \bar{\hat{P}}}{\sigma_{\hat{P}}} \right)$$

where $\bar{P}$ and σ_P are mean and variance of corresponding popularity set. Different from SR, the MAE pays more attention on absolute prediction error between real and predicted popular score on each single sample. The calculation of MAE is as follows:

$$MAE = \frac{1}{N} \sum_{i=1}^N |\hat{P}_i - P_i|$$

4.3 Overall Results

To validate the effective of the proposed multi-modal features in social media popular prediction, two popular base models are adopted in our experiments, including LightGBM[16] and CatBoost[27]. The overall experiments are shown in Table 1. Without any special instructions, our experiments in the following part are based on K-Fold cross validation due to its better performance in valid set, which has also been shown in previous work [18].

First of all, the official post information [40] are adopted as basic features in our experiments. We find that CatBoost model could get much better results than LightGBM with standard official features, e.g. CatBoost improves LightGBM with 0.01 on MAE and 0.049 on SR. Thus, in the following part, the CatBoost are adopted as the default model.

Besides the official post information, similar as HyFea [18], the additional user information has been crawled, e.g. the total view count per each user. It has shown that extra user knowledge could improve the MAE metric with 0.03, which has shown its effectiveness. This is also consistent with our prior that the professional sharers who has many followers and views, could more likely publish posts that attracts people.

Then, the image-related features are explored on popular prediction task. Four image features are adopted in this task, including embedding from the model which is pretrained on the large-scale image-text dataset, the public scene recognition model based on Place365, and two extra image quality assessment models. After adding the collected image features, the MAE and SR could improve with 0.012 and 0.004. This is consistent with our observation that posts that has high image quality, e.g. more aesthetic, would typical better attract people than those posts that has low image quality.

Finally, three text features are extracted to better explore the text semantic knowledge, including prompt text vector from our large-scale image-text multimodal model, the open CLIP model pretrained on 400M image-text pairs, and the BERT model pretrained on large-scale English corpus. As shown in Table 1, after add the text vector features, the MAE and SR could get consistent improvements. The final model with BERT text vector could get 0.1 improvements on

both MAE and SR, than the image-feature based model. We attribute the improvement to the strong semantic representation ability of CLIP/BERT style models, which have been trained on large-scale image-text or language-based datasets.

Table 1: The Overall results with divergence features.

Model	MAE	Sperman's Rho
LightGBM+Official	1.4022	0.6501
CatBoost+Official	1.3913	0.6996
+User Info.	1.3612	0.7009
+Image (Embed./Place365/IQA/NiMa)	1.3491	0.7049
+Prompt text vector	1.2788	0.7091
+Clip text vector	1.2683	0.7142
+Bert text vector	1.2569	0.7144

4.4 Challenge results

Here we have shown our results on SMP 2022 challenge in Table 2. In the final submission to the challenge, seven models which utilize user feature, multi-modal image feature, clip-based text feature are utilized and the average pooling result has been reported. Some better single model results are not included in our final submission due to time issue. In our experiments, we find that it is harder to improve SR than MAE. We guess that it is due to the optimization objective that which has only MSE loss. In the future, the ranking loss will be explored to improve the overall results.

Table 2: SMP 2022 challeng results.

Model	MAE	Sperman's Rho
Canl	1.42	0.69
Jiageng	1.40	0.70
Encu_aida[18]	1.38	0.70
UESTC IntelliGame Lab	1.27	0.71
DMFirstTry(ours)	1.24	0.71

5 CONCLUSION

In this paper, we have proposed a method called "DEVL" to deeply exploit visual and language information for predicting the popularity of social media content. We extract rich features by using several pretrained models from the visual image and language texts from user posts on social media platform, and aim to explore their influence to the social media attractiveness or popularity. The deeply exploited features provides more valuable information than hand-craft categorical features or other deep learning features only trained from social media dataset. Specifically, images are parsed by computing image assessment score, semantic embedding and scene analysis. Then, we extract different multiple semantic embedding from all available post texts including title, tags, concept and category. With the rich visual and language information, it makes social media popularity prediction more accurate. We conduct several experiments on the Social Media Prediction Dataset (SMPD), and achieve new state-of-the-art results on the SMP Challenge at ACM Multimedia 2022.

REFERENCES

- [1] Max Bain, Arsha Nagrani, Gül Varol, and Andrew Zisserman. 2022. A CLIP-Hitchhiker's Guide to Long Video Retrieval. *arXiv preprint arXiv:2205.08508* (2022).
- [2] Qi Cao, Huawei Shen, Jinhua Gao, Bingzheng Wei, and Xueqi Cheng. 2020. Popularity prediction on social platforms with coupled graph neural networks. In *Proceedings of the 13th International Conference on Web Search and Data Mining*. 70–78.
- [3] Guandan Chen, Qingchao Kong, Nan Xu, and Wenji Mao. 2019. NPP: A neural popularity prediction model for social media content. *Neurocomputing* 333 (2019), 221–230.
- [4] Junhong Chen, Dayong Liang, Zhanmo Zhu, Xiaojing Zhou, Zihan Ye, and Xiuyun Mo. 2019. Social media popularity prediction based on visual-textual features with XGBoost. In *Proceedings of the 27th ACM International Conference on Multimedia*. 2692–2696.
- [5] Justin Cheng, Lada Adamic, P Alex Dow, Jon Michael Kleinberg, and Jure Leskovec. 2014. Can cascades be predicted?. In *Proceedings of the 23rd international conference on World wide web*. 925–936.
- [6] Jaemin Cho, Seunghyun Yoon, Ajinkya Kale, Franck Dernoncourt, Trung Bui, and Mohit Bansal. 2022. Fine-grained image captioning with clip reward. *arXiv preprint arXiv:2205.13115* (2022).
- [7] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2018. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805* (2018).
- [8] Keyan Ding, Ronggang Wang, and Shiqi Wang. 2019. Social media popularity prediction: A multiple feature fusion approach with deep neural networks. In *Proceedings of the 27th ACM International Conference on Multimedia*. 2682–2686.
- [9] Emilio Ferrara, Roberto Interdonato, and Andrea Tagarelli. 2014. Online Popularity and Topical Interests Through the Lens of Instagram. In *Proceedings of the 25th ACM Conference on Hypertext and Social Media* (Santiago, Chile) (HT '14). 24–34.
- [10] Francesco Gelli, Tiberio Uricchio, Marco Bertini, Alberto Del Bimbo, and Shih-Fu Chang. 2015. Image popularity prediction in social media using sentiment and context features. In *Proceedings of the 23rd ACM international conference on Multimedia*. 907–910.
- [11] Xiuye Gu, Tsung-Yi Lin, Weicheng Kuo, and Yin Cui. 2021. Open-vocabulary Object Detection via Vision and Language Knowledge Distillation. *arXiv preprint arXiv:2104.13921* (2021).
- [12] Shintami C Hidayati, Kai-Lung Hua, Wen-Huang Cheng, and Shih-Wei Sun. 2014. What are the Fashion Trends in New York. In *Proceedings of ACM International Conference on Multimedia* (ACM MM).
- [13] Bogdan Ionescu, Maia Rohm, Bogdan Boteanu, Alexandru-Lucian Ginsca, Mihai Lupu, and Henning Müller. 2021. Benchmarking Image Retrieval Diversification Techniques for Social Media. *IEEE TMM* 23 (2021), 677–691. <https://doi.org/10.1109/TMM.2020.2986579>
- [14] Chao Jia, Yinfei Yang, Ye Xia, Yi-Ting Chen, Zarana Parekh, Hieu Pham, Quoc Le, Yun-Hsuan Sung, Zhen Li, and Tom Duerig. 2021. Scaling up visual and vision-language representation learning with noisy text supervision. In *International Conference on Machine Learning*. PMLR, 4904–4916.
- [15] Peipei Kang, Zehang Lin, Shaohua Teng, Guipeng Zhang, Lingni Guo, and Wei Zhang. 2019. Catboost-based framework with additional user information for social media popularity prediction. In *Proceedings of the 27th ACM international conference on multimedia*. 2677–2681.
- [16] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. 2017. Lightgbm: A highly efficient gradient boosting decision tree. *Advances in neural information processing systems* 30 (2017).
- [17] Aditya Khosla, Atish Das Sarma, and Raffay Hamid. 2014. What Makes an Image Popular?. In *Proceedings of International World Wide Web Conference (WWW)*.
- [18] Xin Lai, Yihong Zhang, and Wei Zhang. 2020. HyFea: Winning Solution to Social Media Popularity Prediction for Multimedia Grand Challenge 2020. In *Proceedings of the 28th ACM International Conference on Multimedia*. 4565–4569.
- [19] Cheng Li, Yue Lu, Qiaozhu Mei, Dong Wang, and Sandeep Pandey. 2015. Click-through Prediction for Advertising in Twitter Timeline. In *Proceedings of KDD*.
- [20] Jiahao Li, Greg Shakhnarovich, and Raymond A Yeh. 2022. Adapting CLIP For Phrase Localization Without Further Training. *arXiv preprint arXiv:2204.03647* (2022).
- [21] Yiyi Li and Ying Xie. 2020. Is a Picture Worth a Thousand Words? An Empirical Study of Image Content and Social Media Engagement. *Journal of Marketing Research* 57, 1 (2020), 1–19. <https://doi.org/10.1177/0022243719881113>
- [22] Ling Lo, Chia-Lin Liu, Rong-An Lin, Bo Wu, and Wen-Huang Cheng. 2019. Dressing for Attention: Outfit Based Fashion Popularity Prediction. In *IEEE International Conference on Image Processing (ICIP)*.
- [23] Huaishao Luo, Lei Ji, Ming Zhong, Yang Chen, Wen Lei, Nan Duan, and Tianrui Li. 2021. Clip4clip: An empirical study of clip for end to end video clip retrieval. *arXiv preprint arXiv:2104.08860* (2021).
- [24] Tao Mei, Bo Yang, Xian-Sheng Hua, and Shipeng Li. 2011. Contextual video recommendation by multimodal relevance and user feedback. *ACM Transactions on Information Systems* (2011).
- [25] Ron Mokady, Amir Hertz, and Amit H Bermano. 2021. Clipcap: Clip prefix for image captioning. *arXiv preprint arXiv:2111.09734* (2021).
- [26] Alex Nichol, Prafulla Dhariwal, Aditya Ramesh, Pranav Shyam, Pamela Mishkin, Bob McGrew, Ilya Sutskever, and Mark Chen. 2021. Glide: Towards photorealistic image generation and editing with text-guided diffusion models. *arXiv preprint arXiv:2112.10741* (2021).
- [27] Liudmila Prokhorenkova, Gleb Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. 2018. CatBoost: unbiased boosting with categorical features. *Advances in neural information processing systems* 31 (2018).
- [28] Xueming Qian, He Feng, Guoshuai Zhao, and Tao Mei. 2014. Personalized Recommendation Combining User Interest and Social Circle. *IEEE Transactions on Knowledge and Data Engineering* 26, 7 (2014), 1763–1777.
- [29] Jiezhong Qiu, Jian Tang, Hao Ma, Yuxiao Dong, Kuansan Wang, and Jie Tang. 2018. Deepinf: Social influence prediction with deep learning. In *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*. 2110–2119.
- [30] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. 2021. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning*. PMLR, 8748–8763.
- [31] Aditya Ramesh, Prafulla Dhariwal, Alex Nichol, Casey Chu, and Mark Chen. 2022. Hierarchical text-conditional image generation with clip latents. *arXiv preprint arXiv:2204.06125* (2022).
- [32] Chitwan Saharia, William Chan, Saurabh Saxena, Lala Li, Jay Whang, Emily Denton, Seyed Kamyar Seyed Ghasemipour, Burcu Karagol Ayan, S Sara Mahdavi, Rapha Gontijo Lopes, et al. 2022. Photorealistic Text-to-Image Diffusion Models with Deep Language Understanding. *arXiv preprint arXiv:2205.11487* (2022).
- [33] Shaolin Su, Qingsen Yan, Yu Zhu, Cheng Zhang, Xin Ge, Jinqui Sun, and Yanning Zhang. 2020. Blindly assess image quality in the wild guided by a self-adaptive hyper network. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 3667–3676.
- [34] Sanjay Subramanian, Will Merrill, Trevor Darrell, Matt Gardner, Sameer Singh, and Anna Rohrbach. 2022. ReCLIP: A Strong Zero-Shot Baseline for Referring Expression Comprehension. *arXiv preprint arXiv:2204.05991* (2022).
- [35] Gabor Szabo and Bernardo A Huberman. 2010. Predicting the popularity of online content. *Commun. ACM* 53, 8 (2010), 80–88.
- [36] Hossein Talebi and Peyman Milanfar. 2018. NIMA: Neural image assessment. *IEEE transactions on image processing* 27, 8 (2018), 3998–4011.
- [37] Alexandru Tatar, Marcelo Dias De Amorim, Serge Fdida, and Panayotis Antoniadis. 2014. A survey on predicting the popularity of web content. *Journal of Internet Services and Applications* 5, 1 (2014), 1–20.
- [38] Kai Wang, Penghui Wang, Xin Chen, Qiushi Huang, Zhendong Mao, and Yongdong Zhang. 2020. A Feature Generalization Framework for Social Media Popularity Prediction. In *Proceedings of the 28th ACM International Conference on Multimedia*. 4570–4574.
- [39] Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Rémi Louf, Morgan Funtowicz, et al. 2019. Huggingface's transformers: State-of-the-art natural language processing. *arXiv preprint arXiv:1910.03771* (2019).
- [40] Bo Wu, Wen-Huang Cheng, Peiye Liu, Bei Liu, Zhaoyang Zeng, and Jiebo Luo. 2019. SMP challenge: An overview of social media prediction challenge 2019. In *Proceedings of the 27th ACM International Conference on Multimedia*. 2667–2671.
- [41] Bo Wu, Wen-Huang Cheng, Yongdong Zhang, and Tao Mei. 2016. Time Matters: Multi-scale Temporalization of Social Media Popularity. In *Proceedings of the 2016 ACM on Multimedia Conference (ACM MM)* (Amsterdam, The Netherlands).
- [42] Chun-Che Wu, Tao Mei, Winston H Hsu, and Yong Rui. 2014. Learning to personalize trending image search suggestion. In *Proceedings of International ACM SIGIR Conference on Research and Development in Information Retrieval*. 727–736.
- [43] Kele Xu, Zhimin Lin, Jianqiao Zhao, Peicang Shi, Wei Deng, and Huaimin Wang. 2020. Multimodal deep learning for social media popularity prediction with attention mechanism. In *Proceedings of the 28th ACM International Conference on Multimedia*. 4580–4584.
- [44] Qingyuan Zhao, Murat A Erdogdu, Hera Y He, Anand Rajaraman, and Jure Leskovec. 2015. Seismic: A self-exciting point process model for predicting tweet popularity. In *Proceedings of ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 1513–1522.
- [45] Bolei Zhou, Agata Lapedriza, Aditya Khosla, Aude Oliva, and Antonio Torralba. 2017. Places: A 10 million image database for scene recognition. *IEEE transactions on pattern analysis and machine intelligence* 40, 6 (2017), 1452–1464.